Machine Learning: Unsupervised Learning

Brendan Tomoschuk

Updates to class schedule

Week 7

- **Monday:** Machine Learning
- **Wednesday**: NO CLASS (Veteran’s day)
- **Friday:** Language Models

Updates to class schedule



Week 7

- **Monday:** Machine Learning
- **Wednesday**: NO CLASS (Veteran's day)
- **Friday:** Language Models

Week 8

- **I am traveling so there will be no new material**
- **Monday:** [remote, recorded] Interview with Dr. Jolene Mork from Microsoft AI
- **Wednesday:** Andrew Cronan - Career Decision Making
- **Friday:** Class is CANCELLED

Finishing up supervised learning…

Both Regression & Classification

Models that can predict

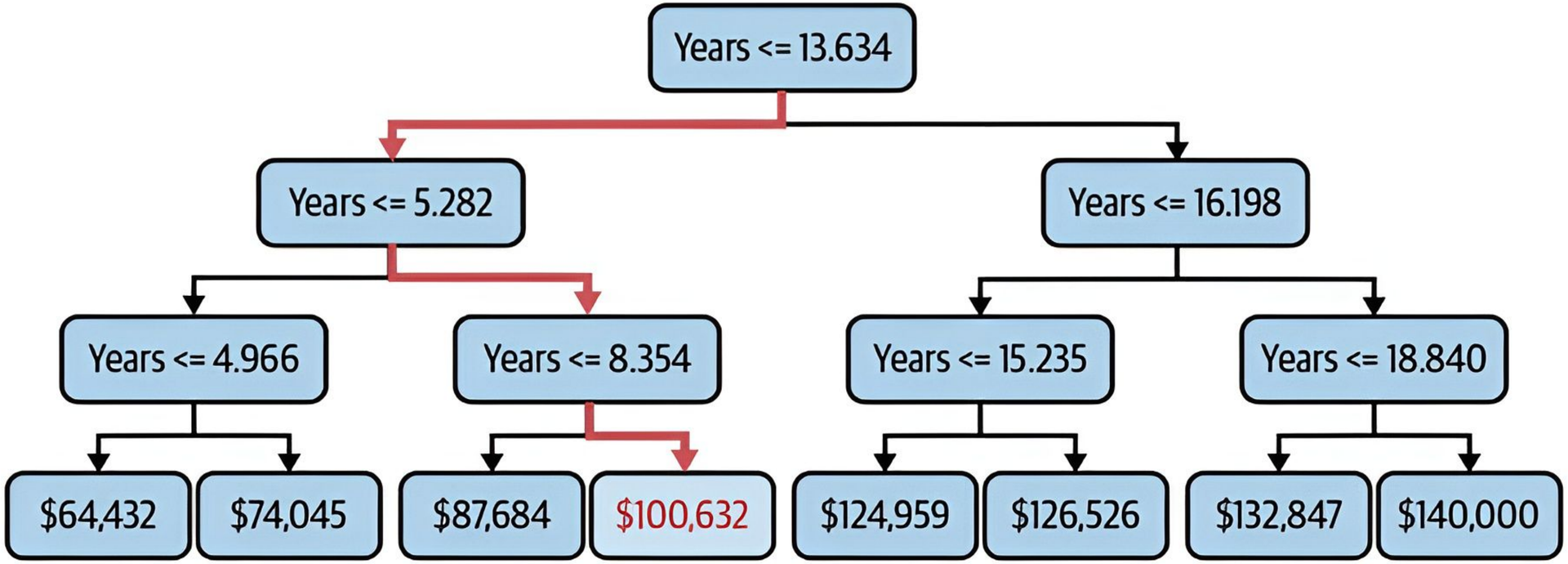
both quantitative and qualitative Y

Decision Trees

Decision Trees: a model that makes predictions by **splitting** the predictors into a series of "branches" that can be summarized in a "tree"-like visual

- Good for modeling ***non-linear*** relationships between X and Y

Regression Decision Tree: Predicting salary from years of experience



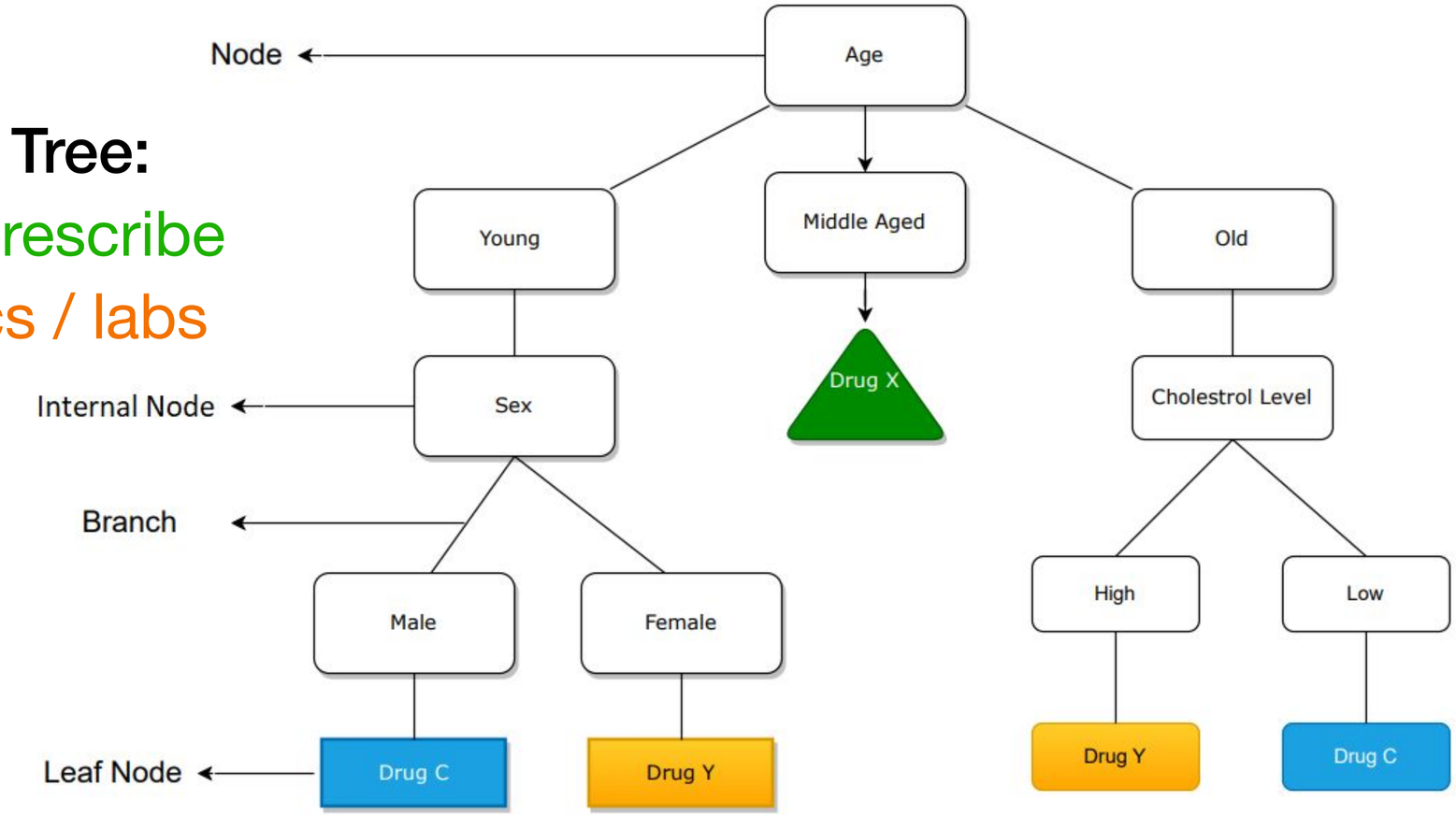
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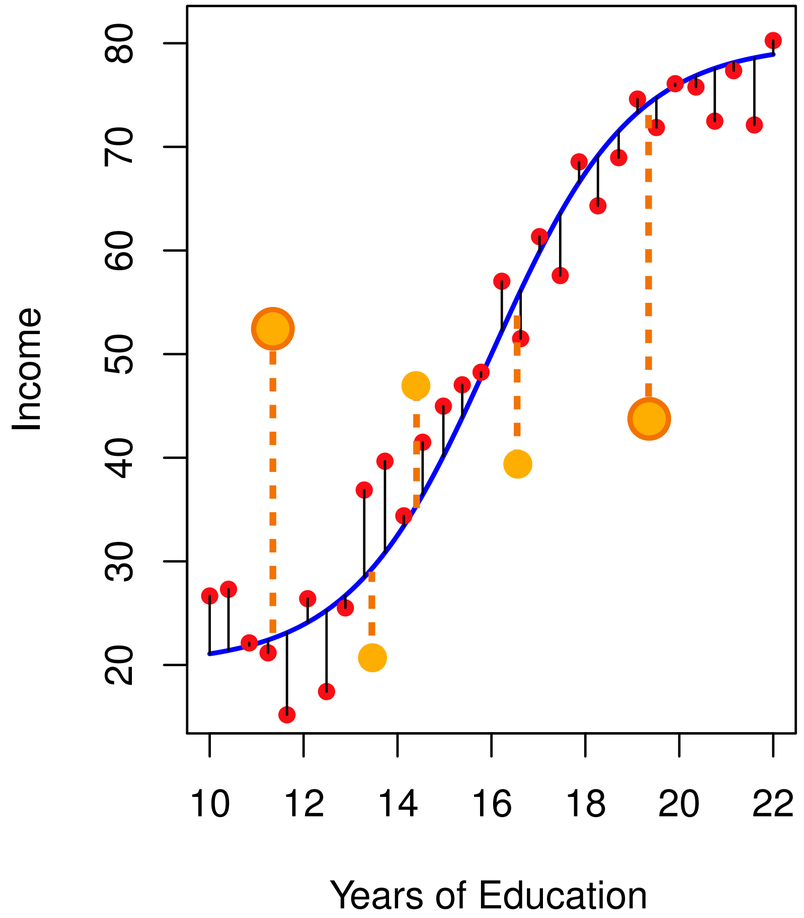
Classification Decision Tree:

Predicting which drug to prescribe from patient demographics / labs



Comparing train and test error for generalization

- You can measure how well the model ***generalizes*** by comparing train and test error
  - **Example:** Predicting someone's income from years of education



$\hat{f}$ estimated from training data

unseen test data

test mean squared error (MSE)

test data with large test MSE

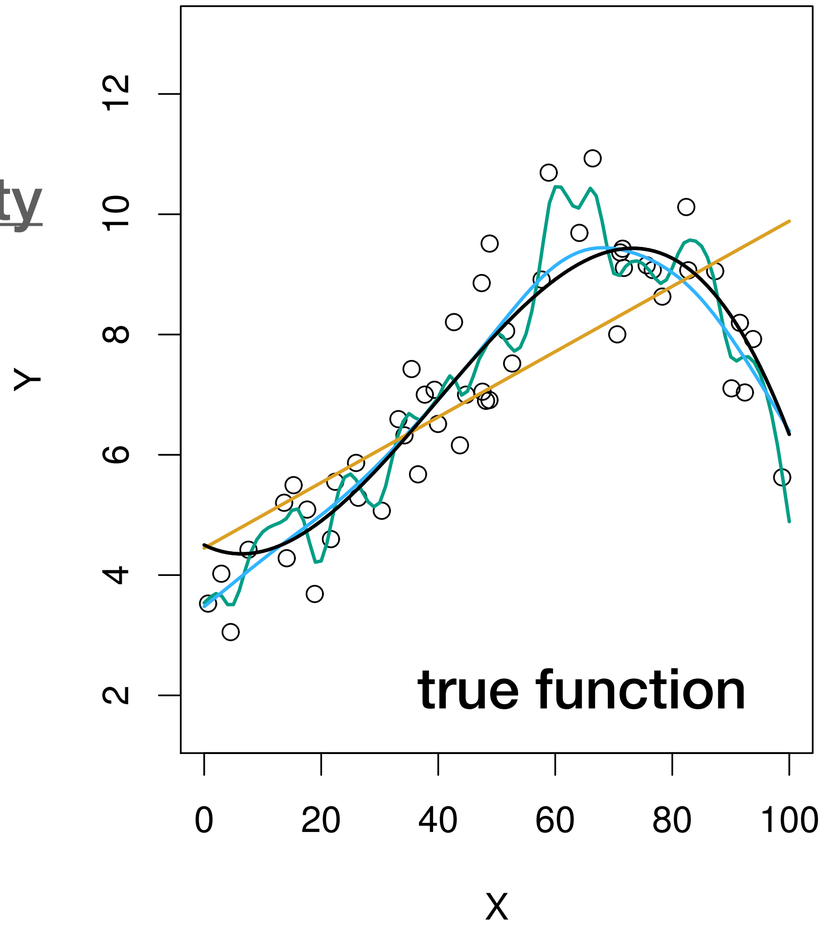
Comparing train and test error for generalization

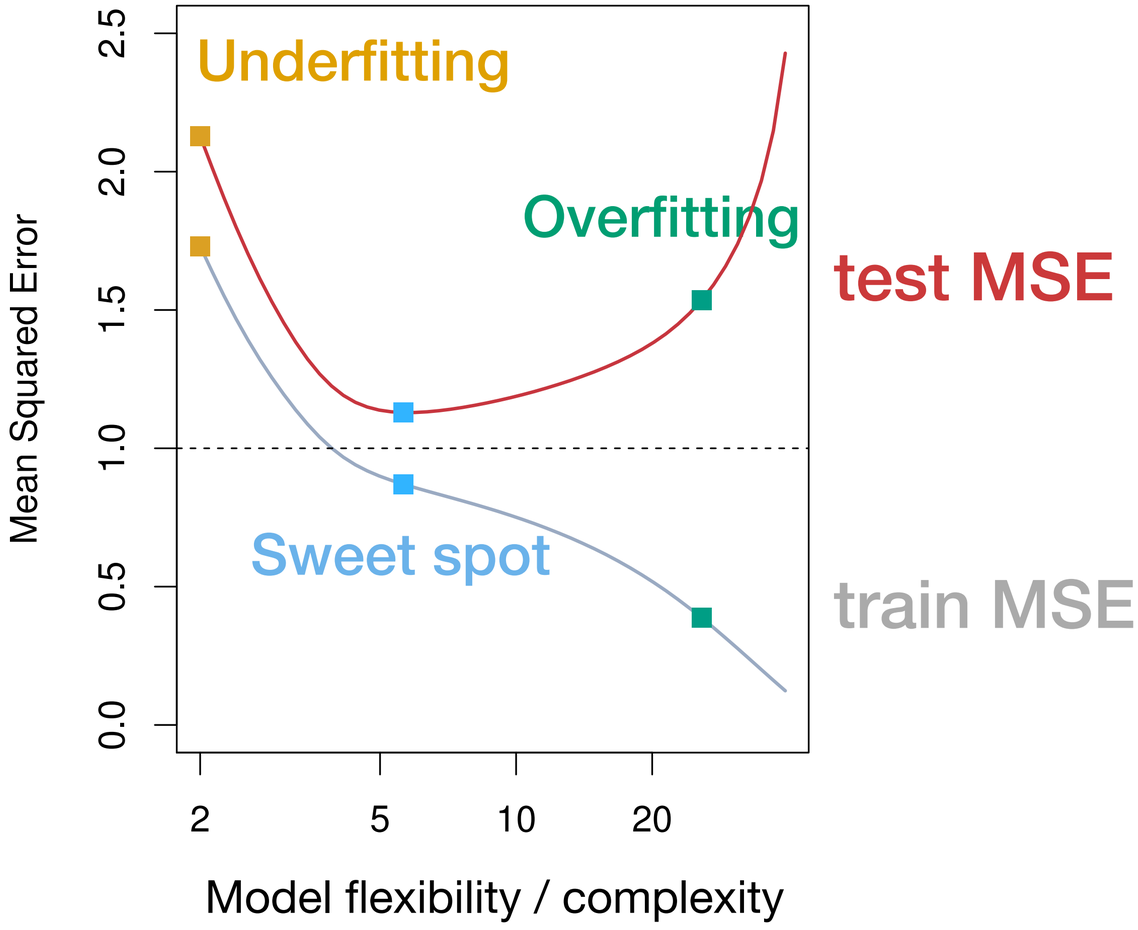
Model flexibility

Least flexible (linear)

Moderate

Most flexible





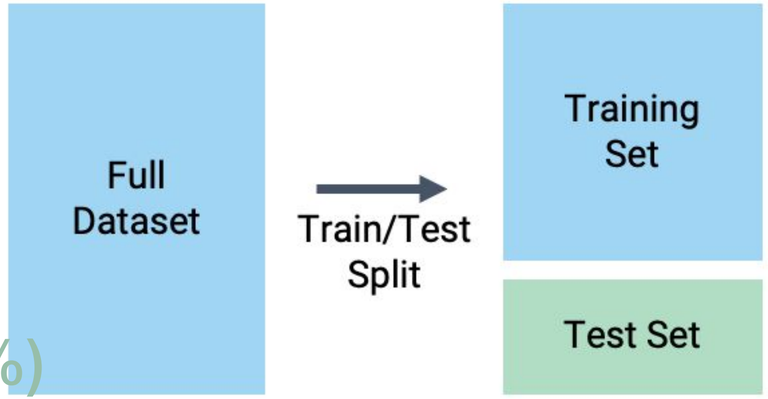
As model flexibility increases, train MSE will always decrease, but test MSE may not

Discuss (~5 minutes)

Your data science question is: Do hours of sleep, major and Spotify listening age predict your numeric grade (0-100) in COGS 9? If so, by how much?

1. What EDA(s) would you choose?
2. What statistical test would you run?
3. What model would you run?
4. Which evaluation metric would you use for your model?

Putting it all together



Supervised learning modeling workflow:

1. Split the data into a training (80%) and a test set (20%)
2. Train the model on the training data
3. Use the trained model to make predictions on the test data
4. Use a model evaluation metric to quantify the model's training and test performance
   - e.g. MSE for **regression** models
   - e.g. accuracy rate or error rate for **classification** models
5. Diagnose overfitting / underfitting by comparing the training and test error
6. Refine the model if needed!

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course code:
WSTAPX

You are using a multiple linear regression model to predict someone's height from their age and weight.

Which model evaluation metric would you use?

(A) Mean squared error

(B) Classification accuracy

(C) True positive rate

(D) False negative rate

(E) I'm not sure / I'm confused

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Which of the following is FALSE about neural networks?

(A) They can capture non-linear relationships between inputs and outputs

(B) LLMs are a type of neural network

(C) Neural networks are classification models

(D) Neural networks are a type of unsupervised learning model

(E) I'm not sure

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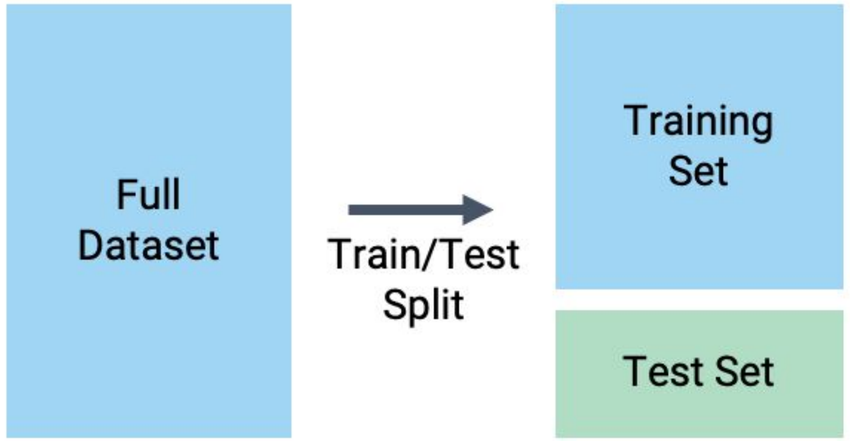
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If a model's training error is high, and the test error is also high, this model is likely…

(A) Underfitting

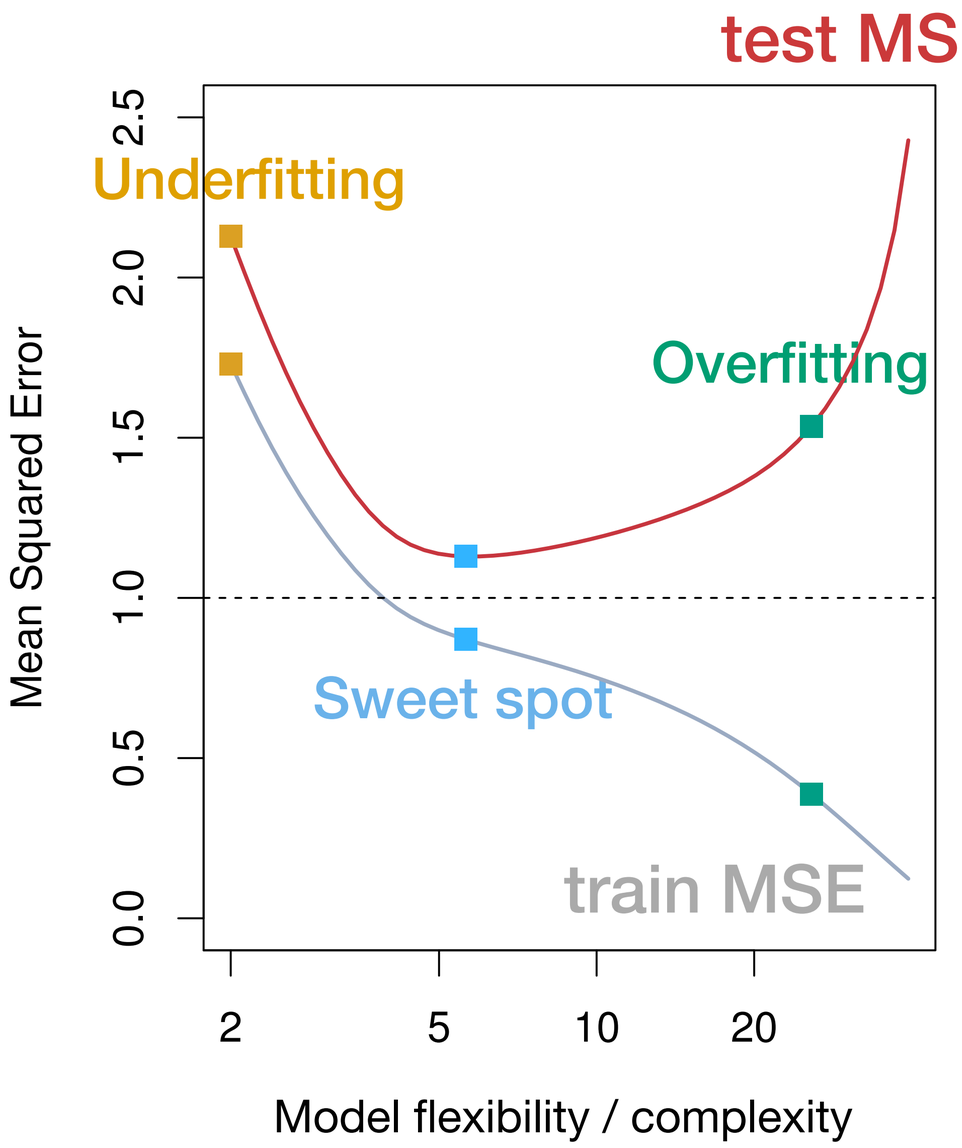
(B) Overfitting

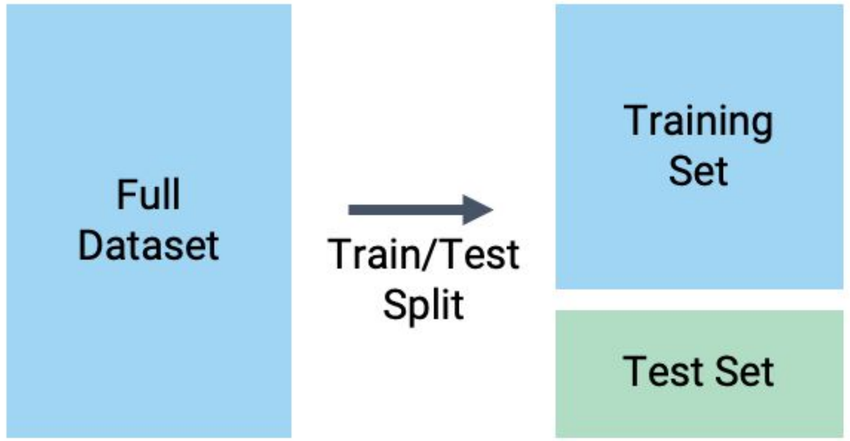
(C) Fits the training data well

(D) Fits the test data well

(E) I'm not sure / I'm confused

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If a model's training error is high, and the test error is also high, this model is likely…

(A) Underfitting

(B) Overfitting

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WSTAPX

You want to predict how long you have to wait for the elevator at Geisel. *What kind of problem is this, and what kind of model evaluation metric would you use?*

(A) Regression; accuracy rate

(B) Classification; accuracy rate

(C) Regression; mean squared error

(D) Classification; mean squared error

(E) I'm not sure / I'm confused

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- For **regression** (quantitative Ys): a common evaluation metric is **mean squared error (MSE)**

Learning Goals

1. Describe unsupervised learning problems and applications
    - Distinguish clustering vs. dimensionality reduction methods
2. Describe reinforcement learning problems and applications

Supervised Learning Models

Supervised Learning: Predicting Y from X

Regression: Predicting quantitative Y

- Linear regression (simple & multiple)
- Non-linear regression

Classification: Predicting qualitative (categorical) Y

- Logistic regression

Both regression & classification:

- Decision trees
- Neural networks

Takeaway: Your choice of model depends on the *outcome* you are predicting as well as the *relationship* between the input and output!

Unsupervised Learning

No labels Y to predict

Machine Learning

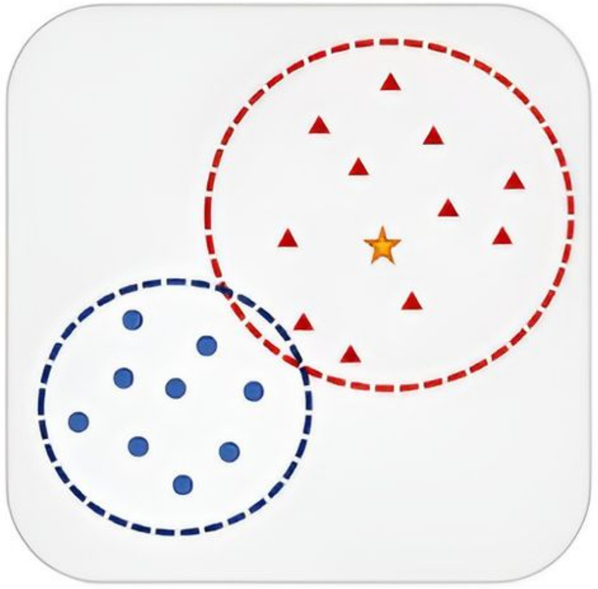
Learning from data to predict or generalize to unseen data

Learning from trial-and-error interactions with the environment

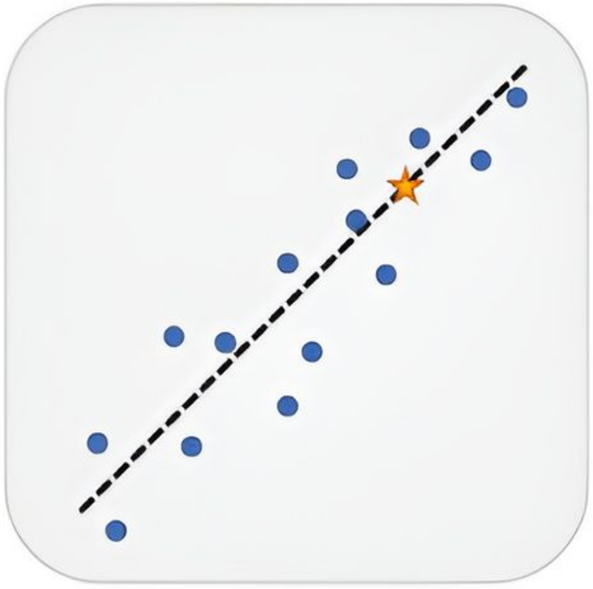
Supervised Learning

Model training with labelled data

Classification



Regression

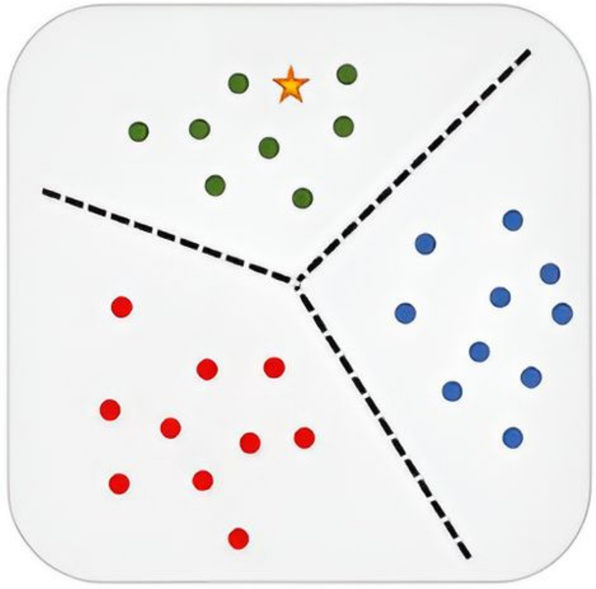


Unsupervised Learning

Model training with unlabelled data

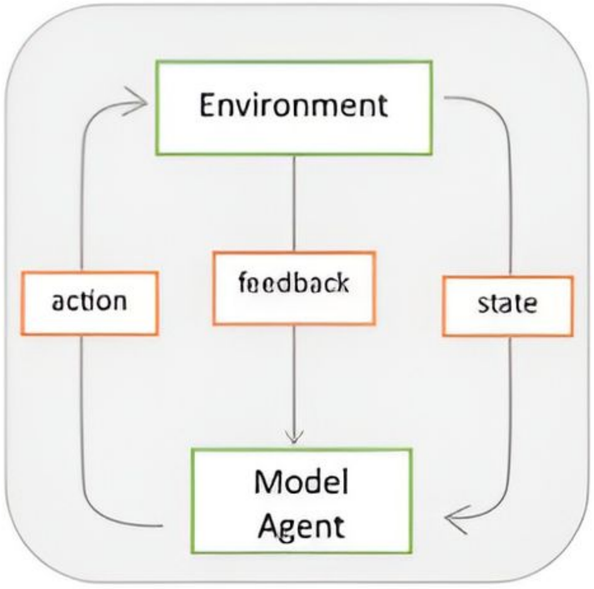
Discovering patterns in data

Clustering



Reinforcement Learning

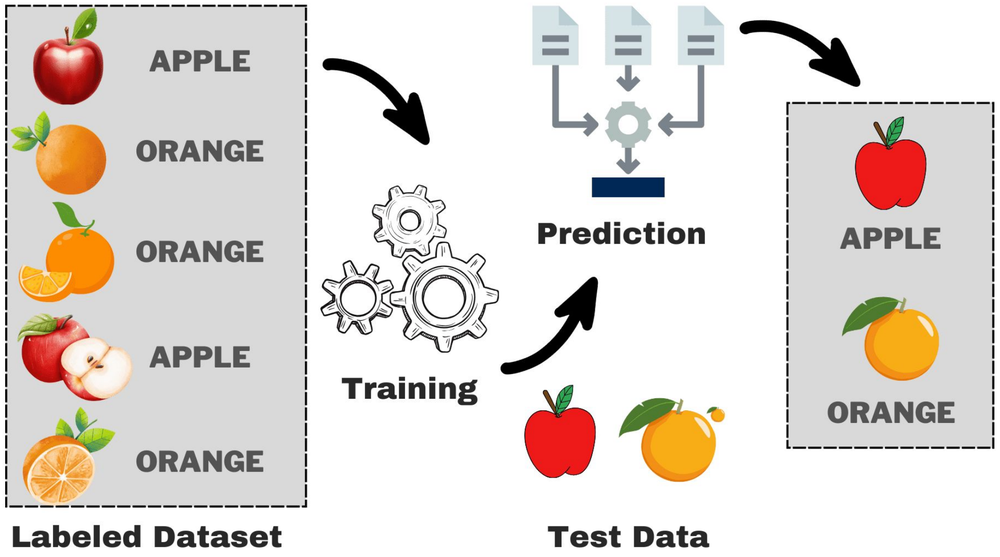
Model take actions in the environment then received state updates and feedbacks



Supervised vs. Unsupervised Learning

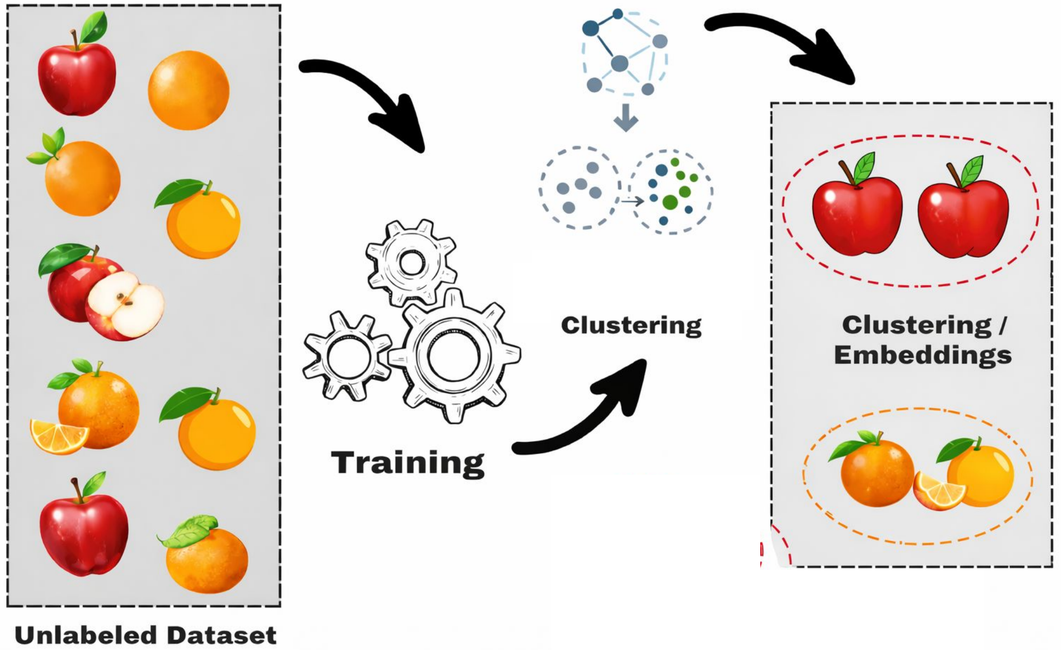
Supervised: there is a correct answer or label to be predicted

- **Goal:** correctly predict labels y_i for each x_i
- Two main types: regression & classification



Unsupervised: no labels or correct answers to predict

- **Goal:** find patterns and structure in unlabeled data by **grouping** or **clustering** data by similarity



Unsupervised Learning

- **Supervised learning**: for every data point x_i, we have a y_i to predict (labels)
- **Unsupervised learning**: models that only require x_is (without associated y_is)
- Dimensionality reduction: Summarizing or visualizing **high-dimensional data** without losing the main information
  - ***Models:*** principal components analysis (PCA), t-SNE, UMAP, factor analysis
- Clustering: Grouping the data by similarity of features
  - ***Models:*** K-means clustering, hierarchical clustering

Dimensionality Reduction

PCA and t-SNE to visualize and reveal structure across subtypes of neurons (brain cells)

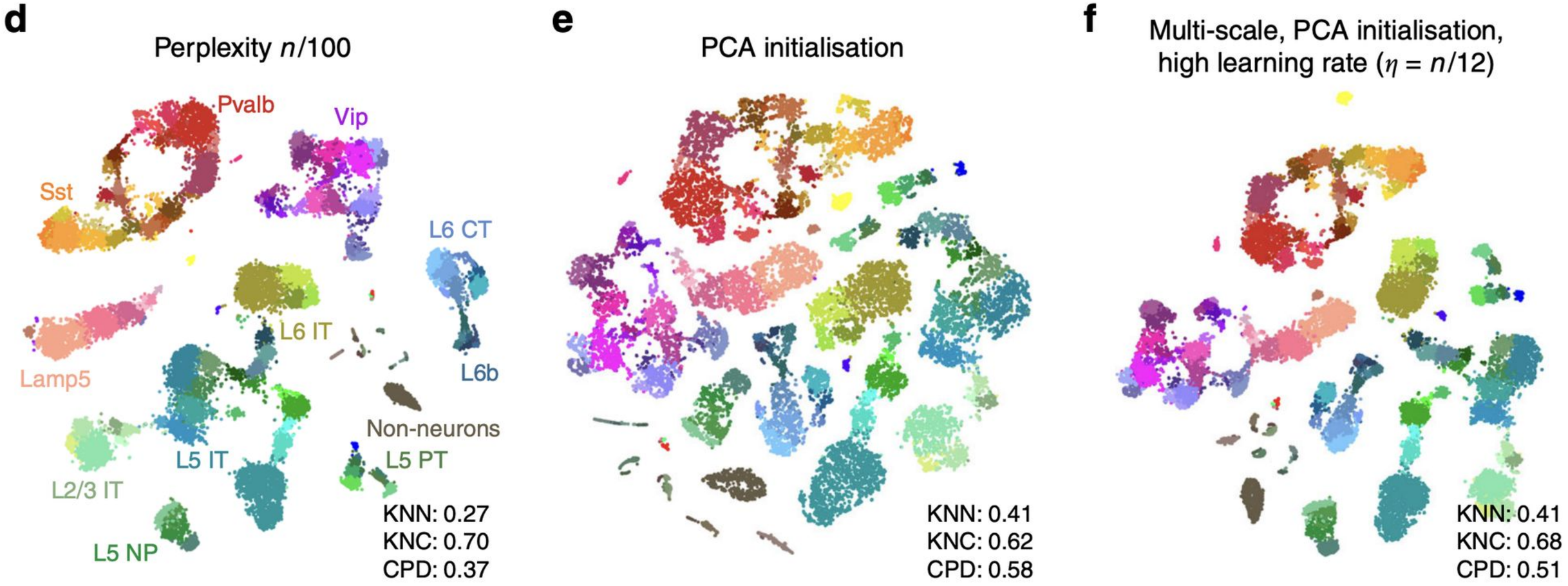
Each variable is a “dimension”→ high-dimensional dataset

neuron_id	PVALB_expr	SST_expr	VIP_expr	SLC17A7_expr	firing_rate_hz	spike_width_ms	…
N01	8.7	0.4	0.2	0.1	28	0.35	
N02	7.9	0.6	0.1	0.2	34	0.30	
N03	9.2	0.3	0.1	0.1	22	0.40	
N04	0.5	8.1	0.3	0.1	12	0.80	
N05	0.3	7.4	0.4	0.2	9	0.95	
N06	0.6	9.0	0.2	0.1	15	0.75	
...							

Dimensionality Reduction

Takes information from many features *(neuron firing, gene expression, connectivity, etc.)* and plots each cell as a dot in a 2-D space so that cell that are very similar are close to each other!

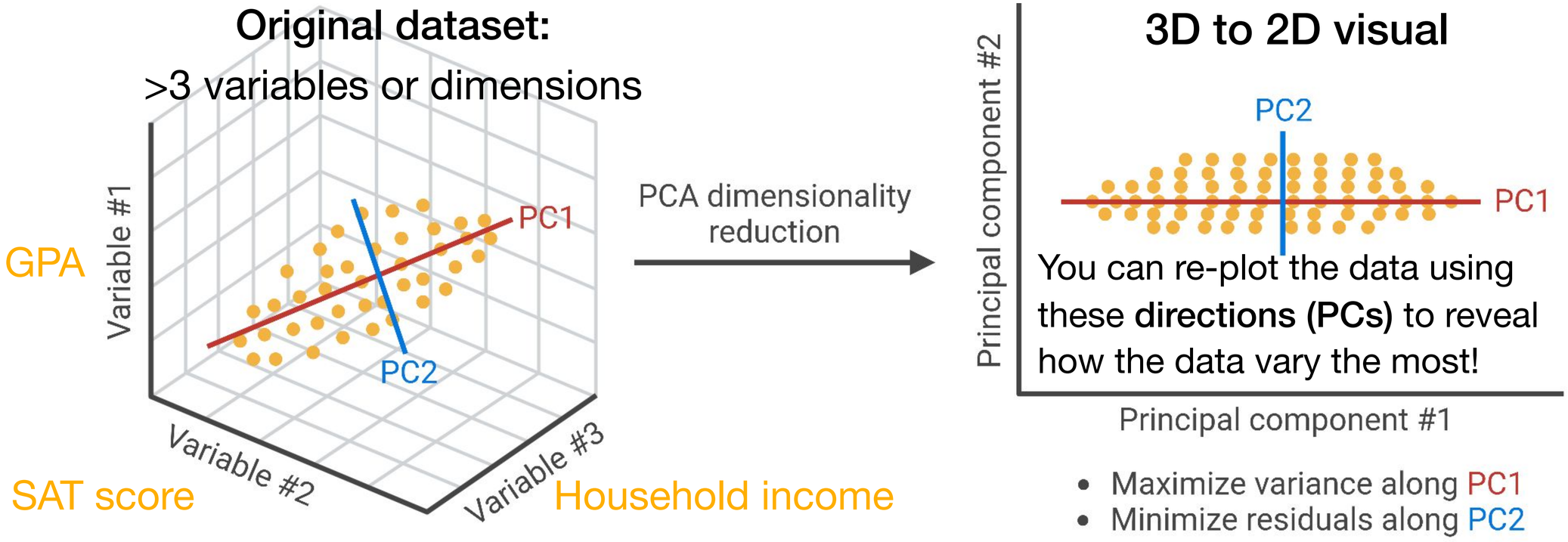
PCA and t-SNE to visualize and reveal structure across subtypes of neurons (brain cells)



Dimensionality Reduction

Principal components analysis (PCA) is a **dimensionality reduction** method for finding a **low-dimensional** representation of the data OR for **visualizing** the data in 2 or 3-D

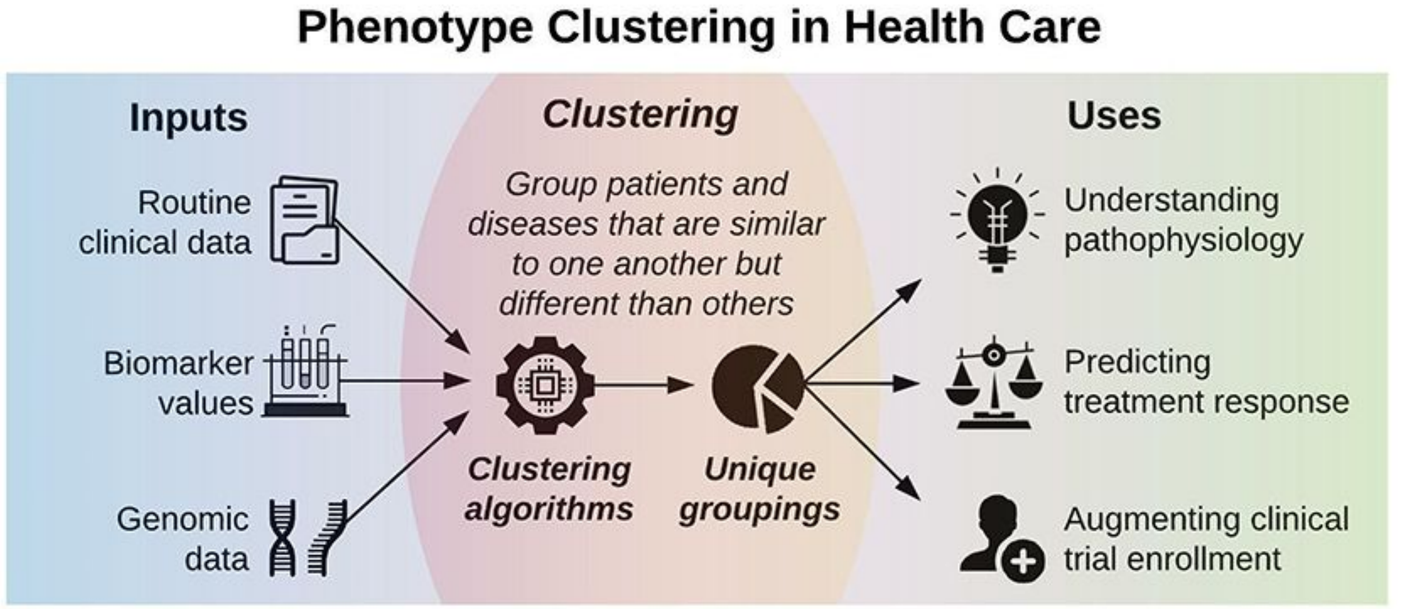
- **Example:** Visualize high-dimensional information about UCSD applicants

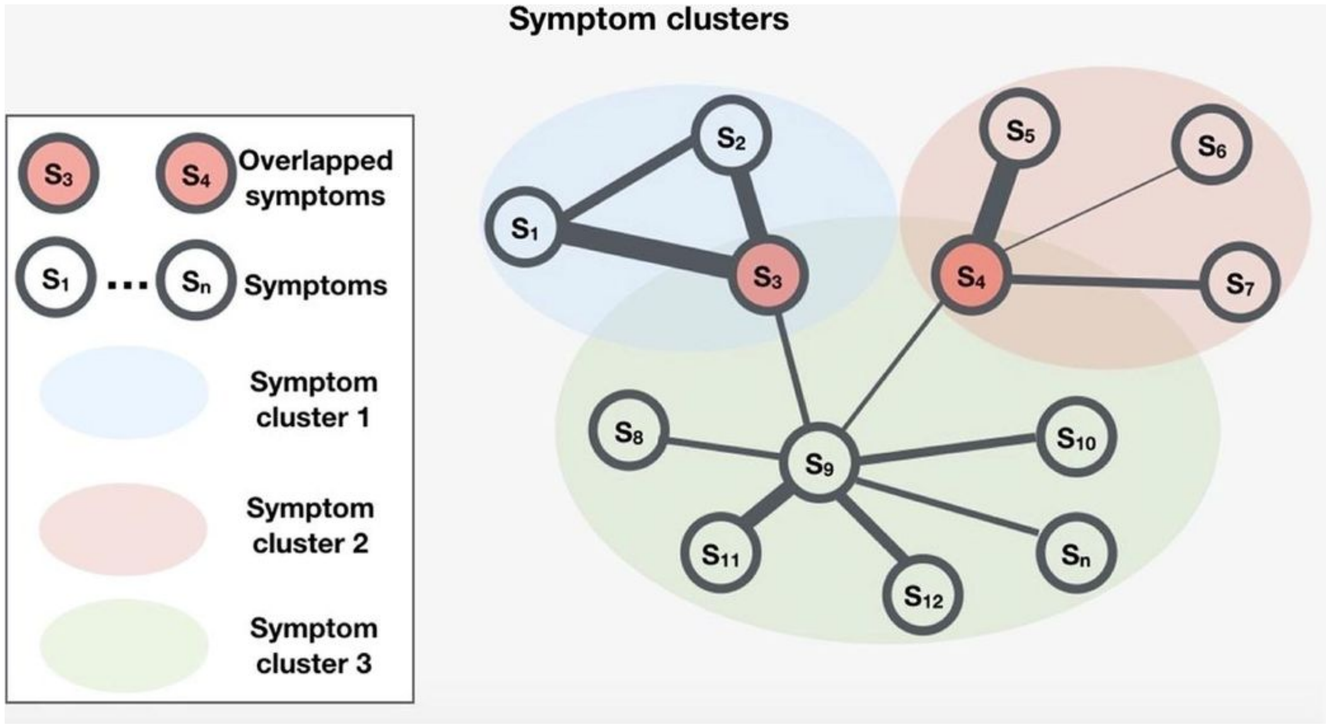


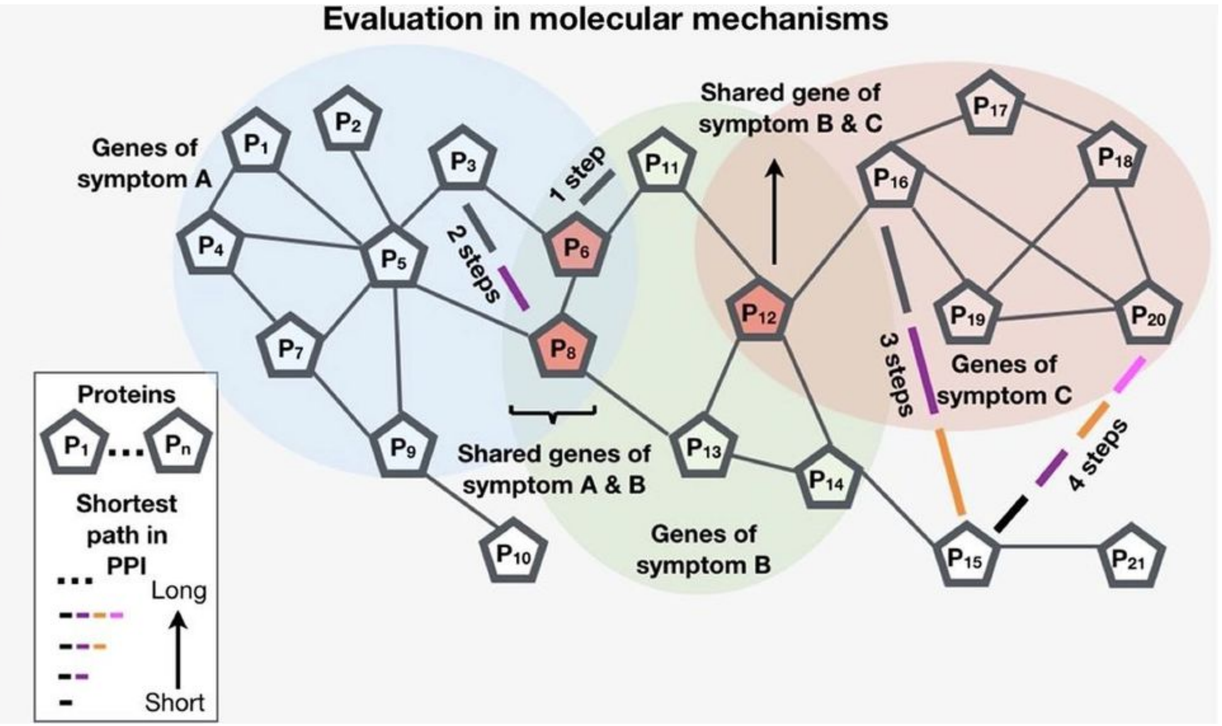
PCA goal: find the **directions (PCs)** in feature space that have the most ***variance***

Clustering

Disease **clustering** by symptoms and genes to understand how disease subgroups might respond to treatments



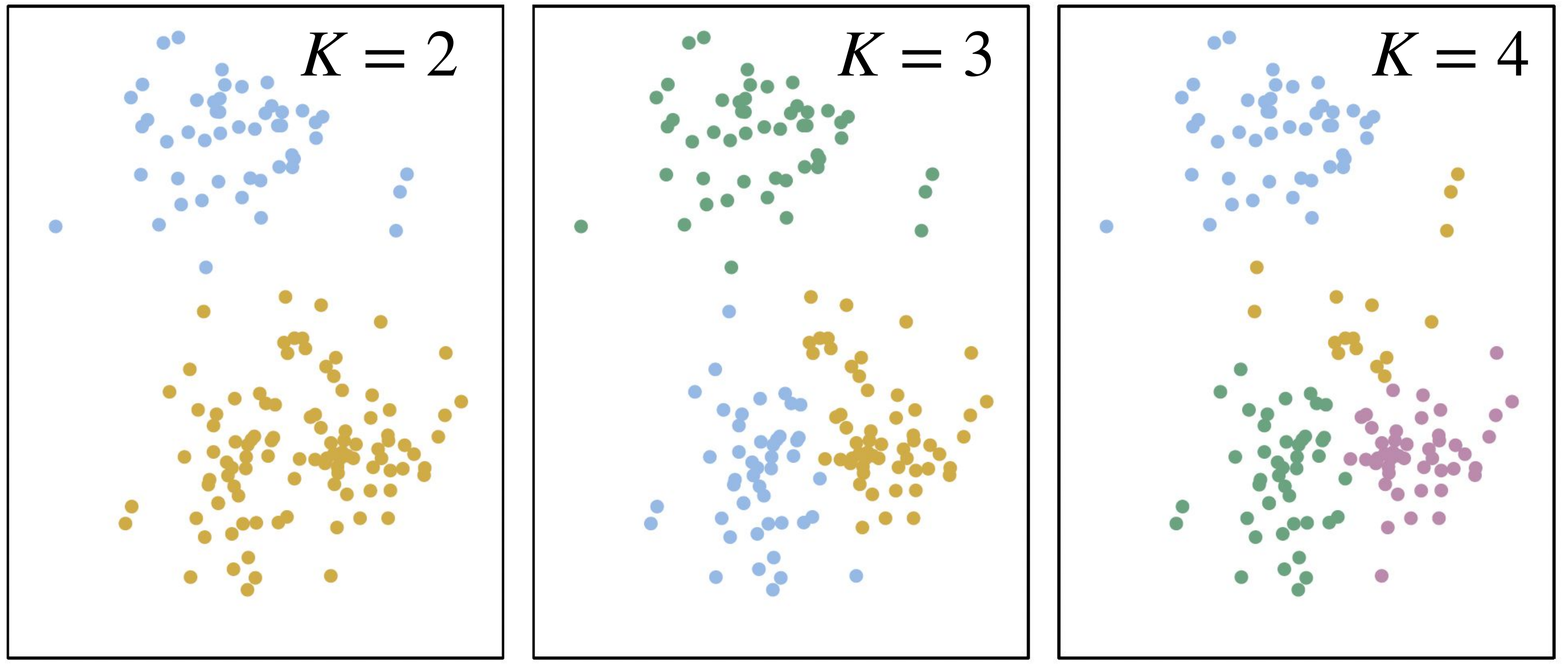




Clustering

K-means clustering groups data points into specified K number of groups based on their features X

- Must choose the value of K in advance
- Color indicates the cluster the datapoint was assigned to:

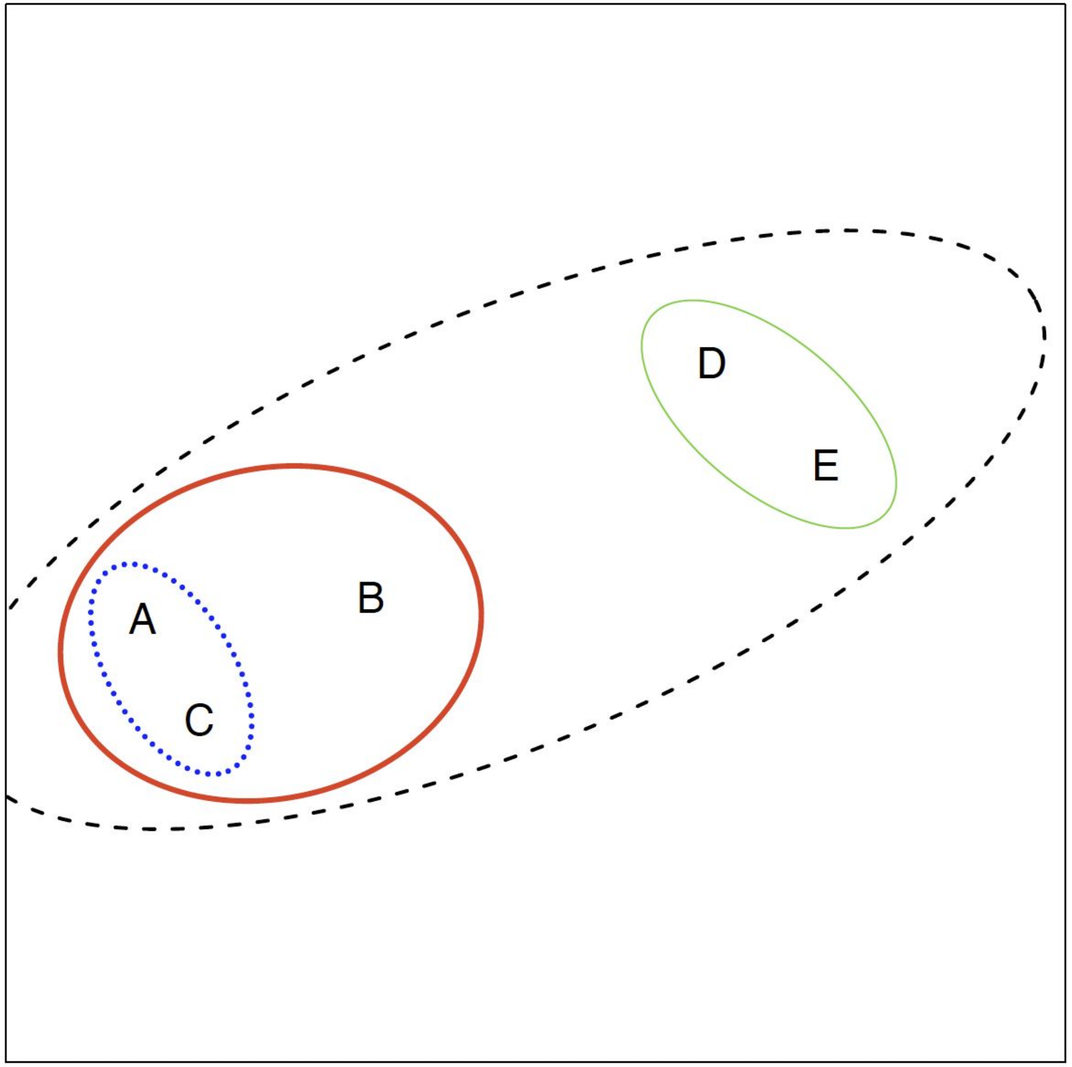


No “right” answer!

Clustering

- *Unlike k-means,* **hierarchical clustering** doesn't require choosing the number of clusters in advance
- **Hierarchical clustering:** Builds a hierarchy of groupings that starts with pairs of data points

X_2: Hours of study

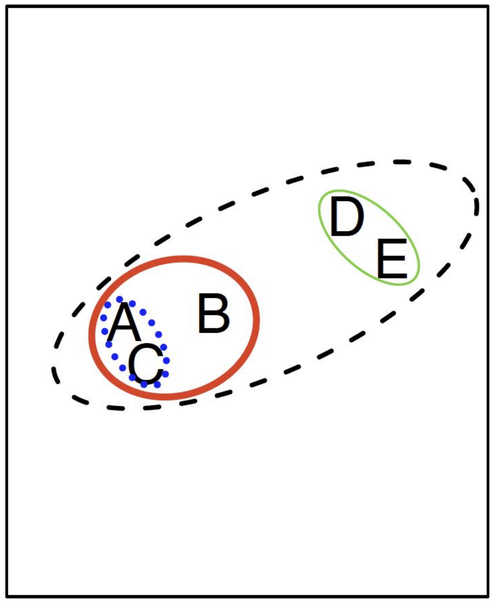


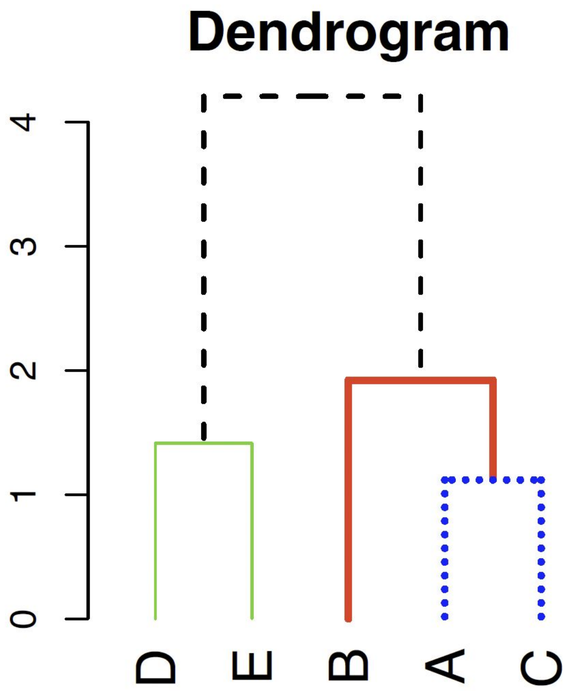
X_1: Hours of sleep

Clustering

Hierarchical clustering algorithm:

1. Start with each data point in its own cluster
2. Identify the **closest two clusters** and merge them
3. Repeat #2 until all points are in a **single** cluster

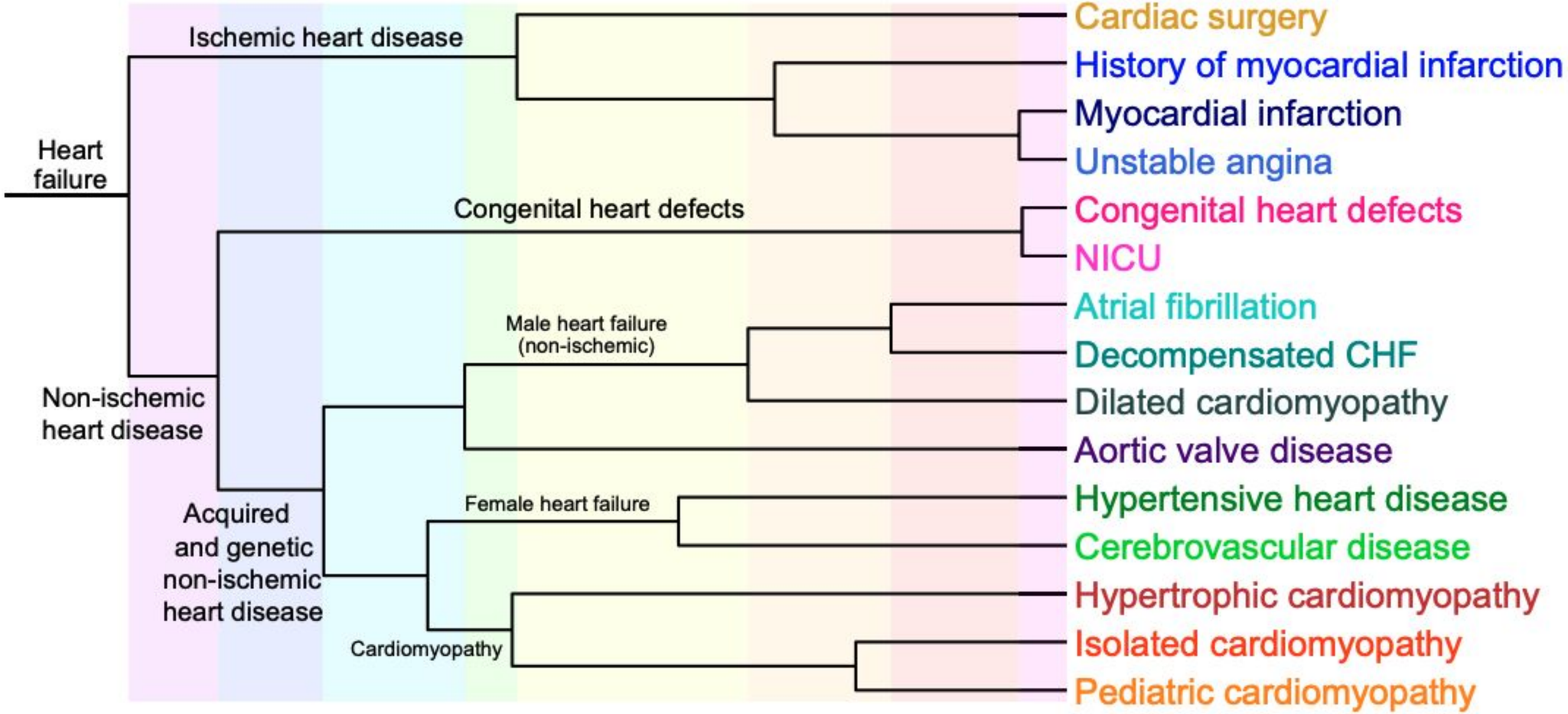




Dendrogram describes similarities between clusters and displays all potential clusters at once

Clustering

Hierarchical clustering for different reasons of heart failure



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You want to visualize some data that has a lot of features (high-dimensional). Which kind of model should you use?

(A) Neural networks

(B) Linear regression

(C) Principal components analysis (PCA)

(D) K-means clustering

(E) I'm not sure / I'm confused

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